IAM-Based Access to EC2 Using IAM Identity Center (Session Manager)

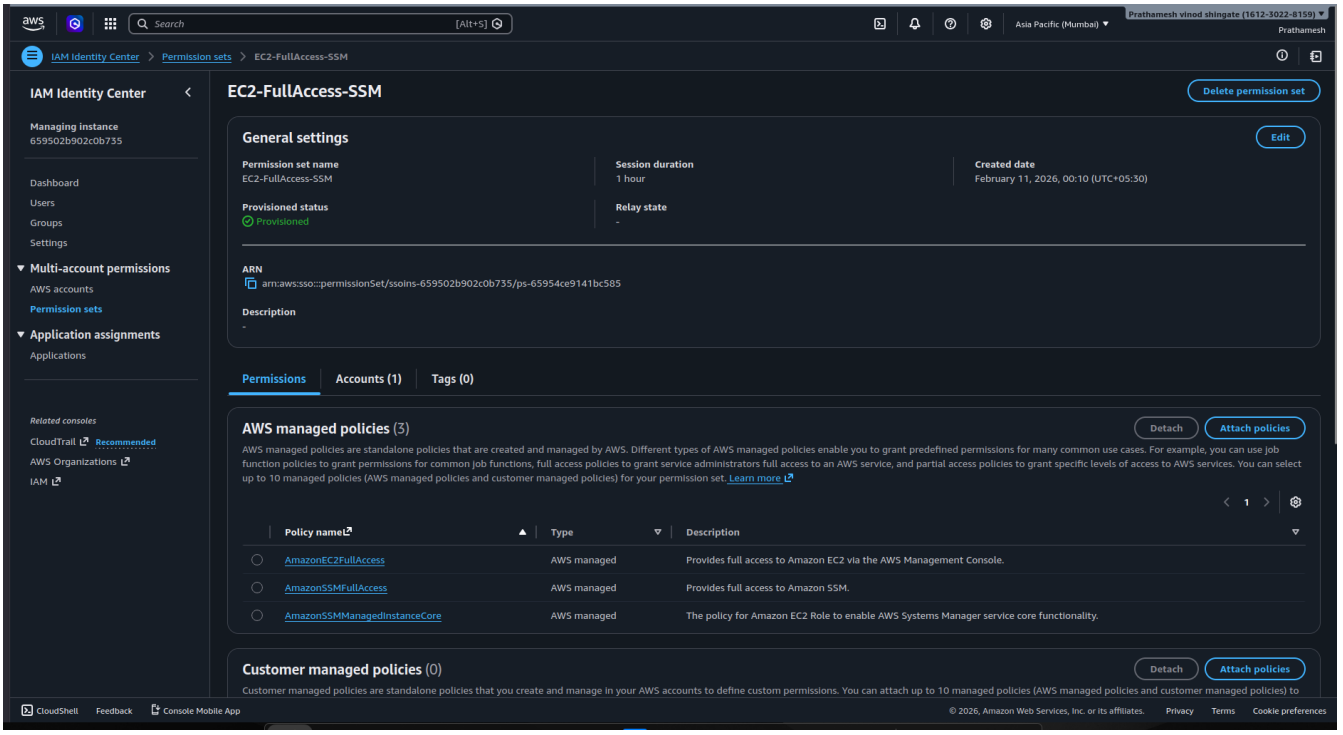
Step 1: Enable IAM Identity Center

First, IAM Identity Center was enabled from the AWS Management Console in ap-south-1(mumbai-region). This was required to manage users without sharing SSH keys.

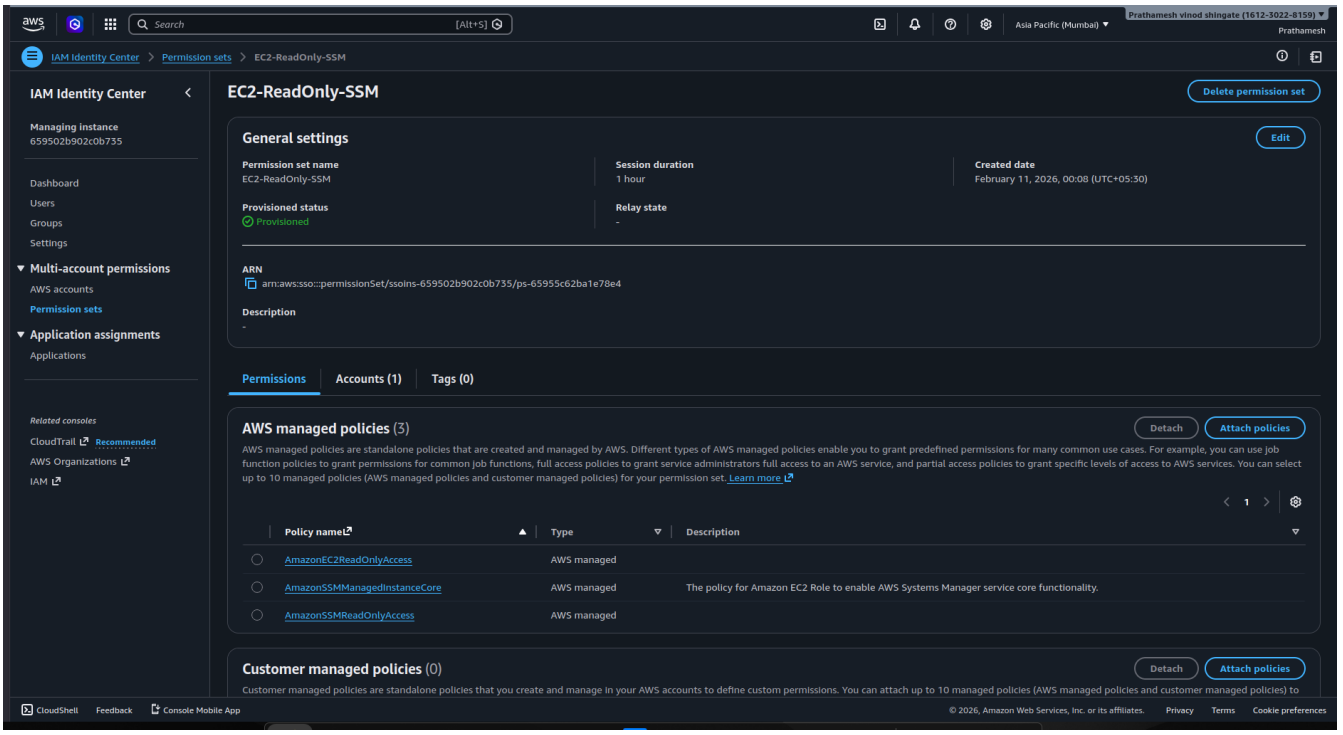
Step 2: Create Permission Sets

Two permission sets were created in IAM Identity Center.

The **first permission set** was created for the DevOps user. This permission set included AmazonEC2FullAccess, AmazonSSMFullAccess and AmazonSSMManagedInstanceCore.



The **second permission set** was created for the Backend user.
This permission set included AmazonEC2ReadOnlyAccess, AmazonSSMReadOnlyAccess and AmazonSSMManagedInstanceCore.

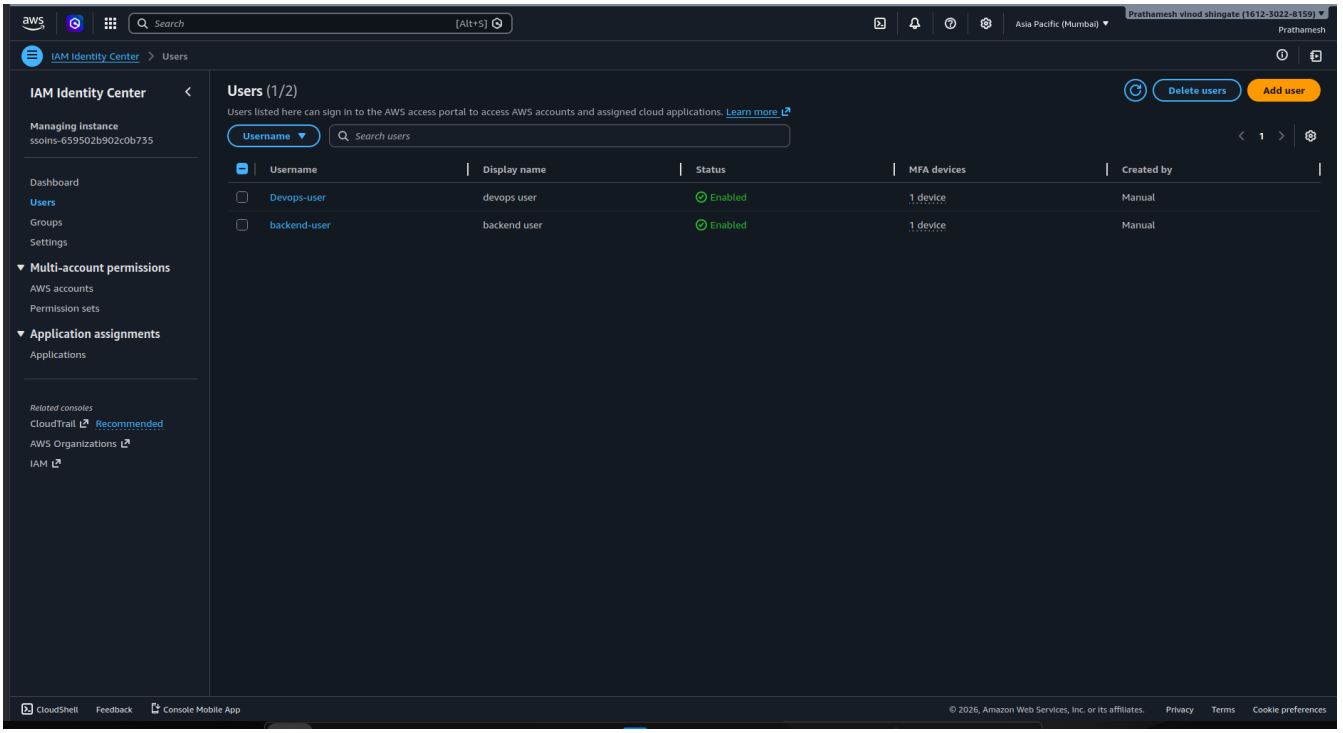


Step 3: Create Users in Identity Center

Two users were created in IAM Identity Center.

One user was created with the name **Devops-user.**
Another user was created with the name **backend-user**.

Each user was created with a unique email address.

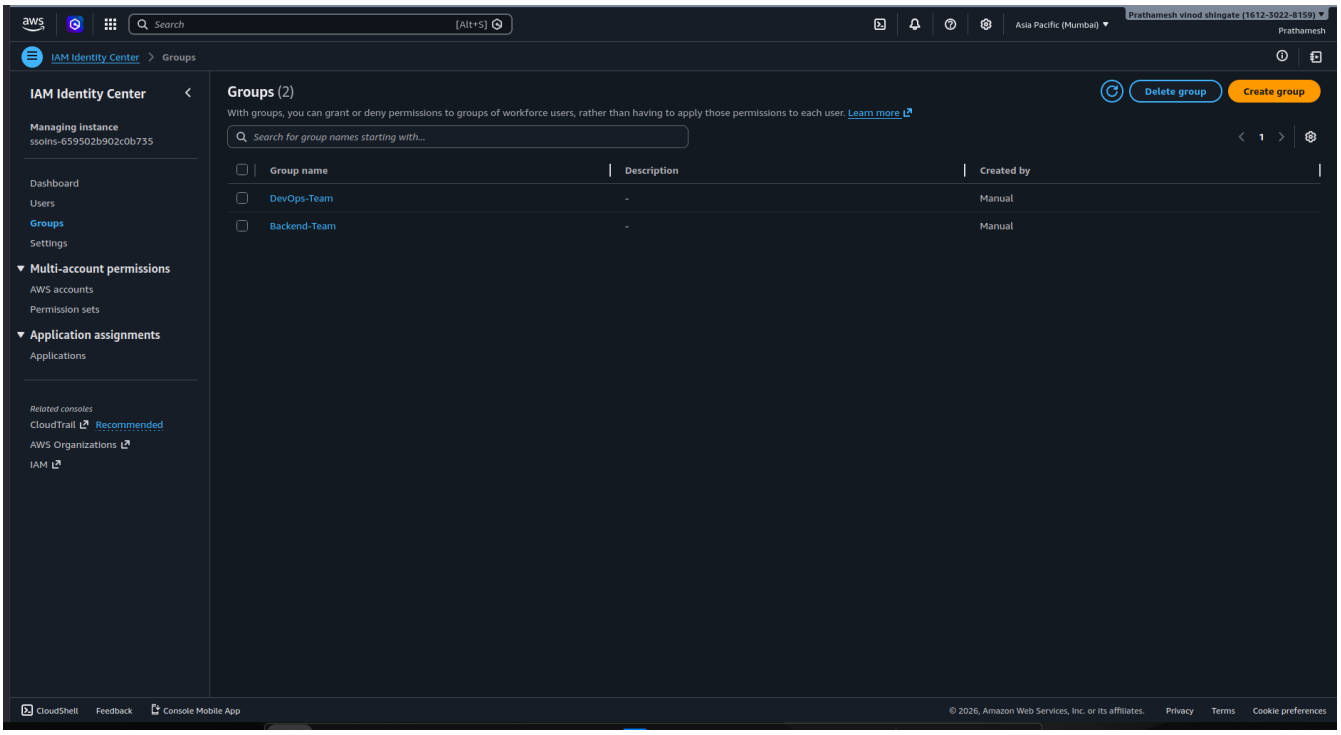


Step 4: Create Groups and Add Users

Two groups were created in IAM Identity Center.

One group was created with the name **DevOps-Team** and the **devops-user** was added to this group.

Another group was created with the name **Backend-Team** and the **backend-user** was added to this group.

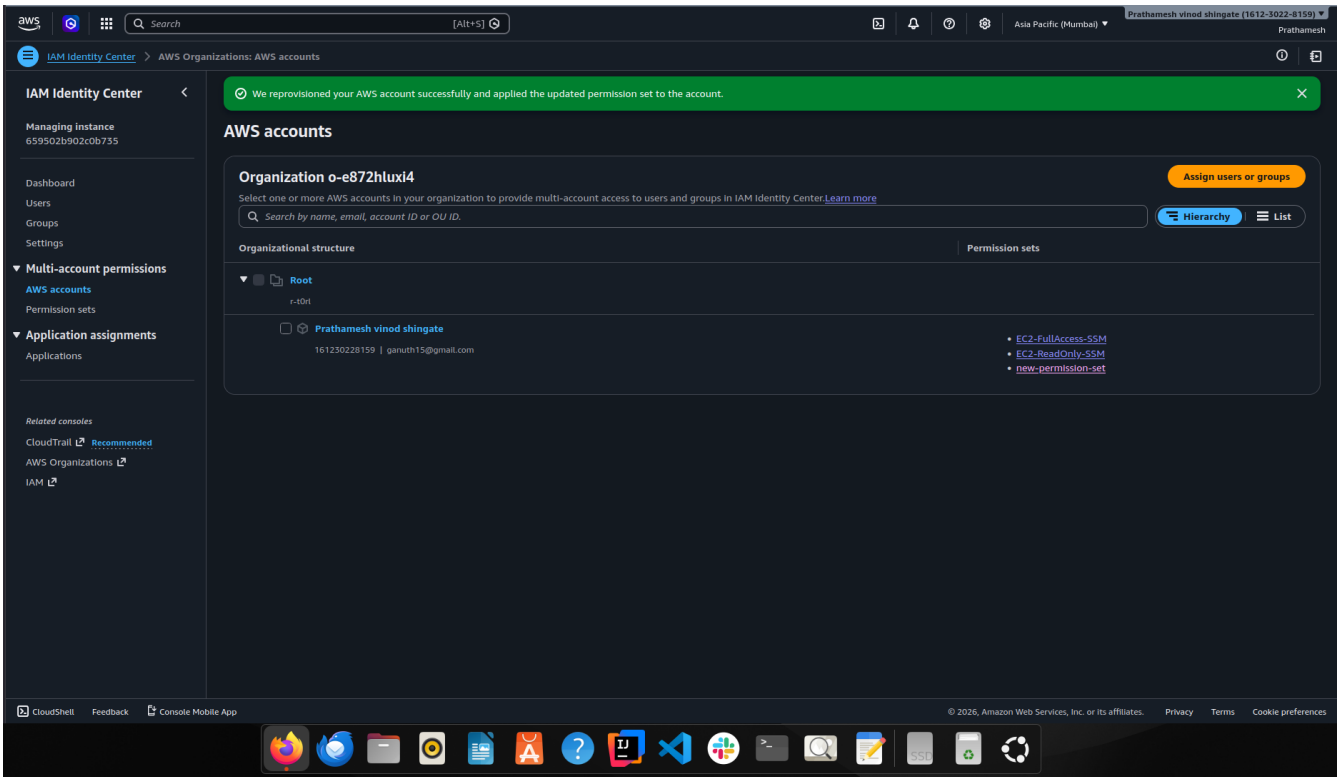


Step 5: Assign Permission Sets

The **EC2-FullAccess-SSM permission set** was assigned to the **DevOps-Team** for the AWS account.

The **EC2-ReadOnly-SSM permission set** was assigned to the **Backend-Team** for the same AWS account.

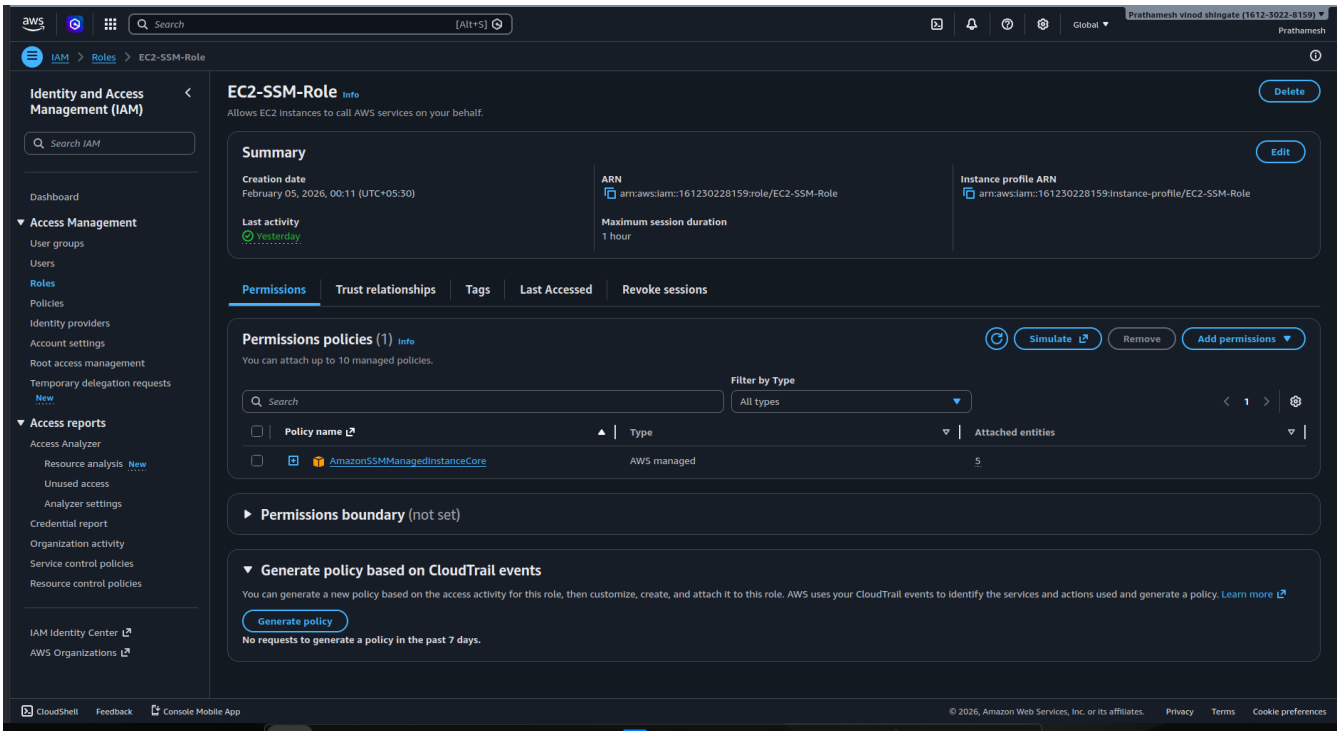
And Also **applied the updated permission set to the Account.**



Step 6: Create IAM Role for EC2 (SSM Role)

An IAM role was created for EC2 with **AmazonSSMManagedInstanceCore** policy attached.

This role allows the EC2 instance to communicate with AWS Systems Manager.



Step 7: Launch EC2 Instance

An AL2 EC2 instance was launched.

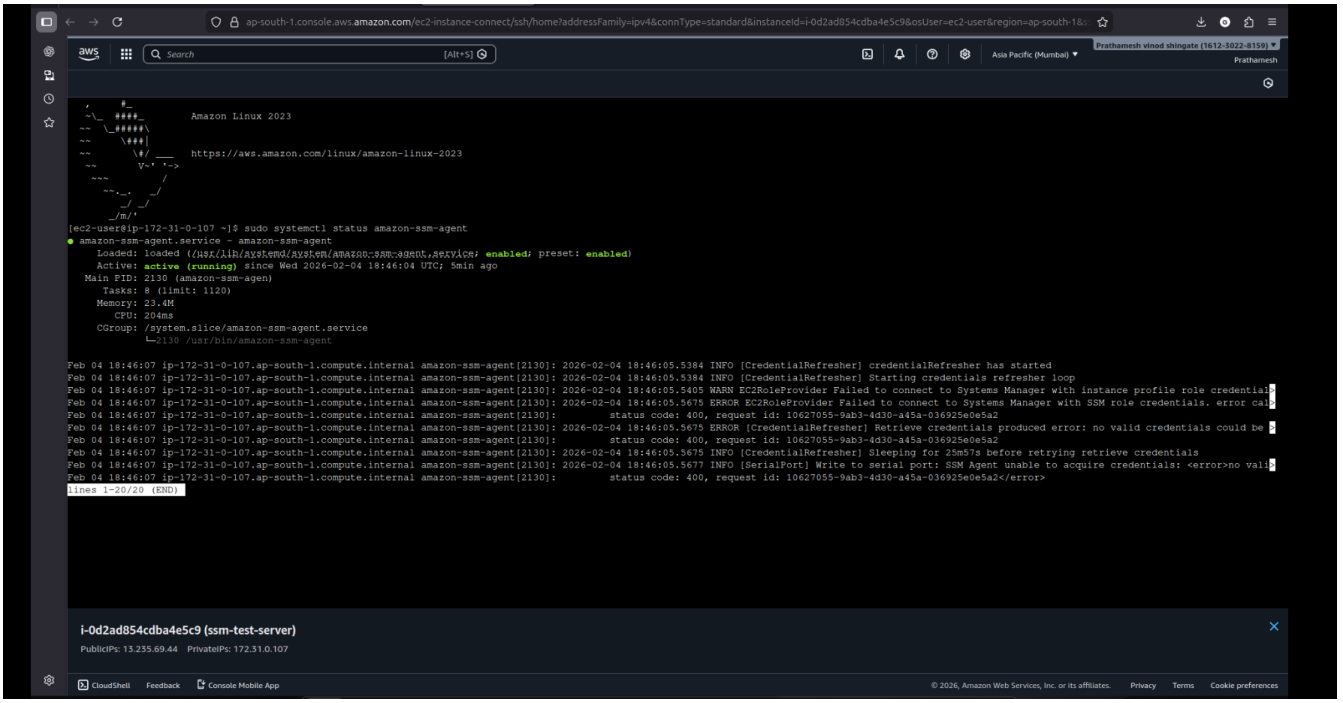
While creating the instance, the SSM **IAM role(EC2-SSM-Role)** was attached to the instance.

Step 8: Install and Start SSM Agent

After the EC2 instance was launched, the instance was accessed using the **admin AWS account** through the **EC2 Connect** option for initial setup.

Once logged in to the AL2 instance, the SSM Agent was installed and the service was started.

The status of the SSM Agent was verified to ensure it was running successfully.



Also we need to restart the amazon-ssm-agent

```
sudo systemctl restart amazon-ssm-agent
```

Step 9: Verify Instance in Fleet Manager

The EC2 instance was checked in AWS Systems Manager → **Fleet Manager**.

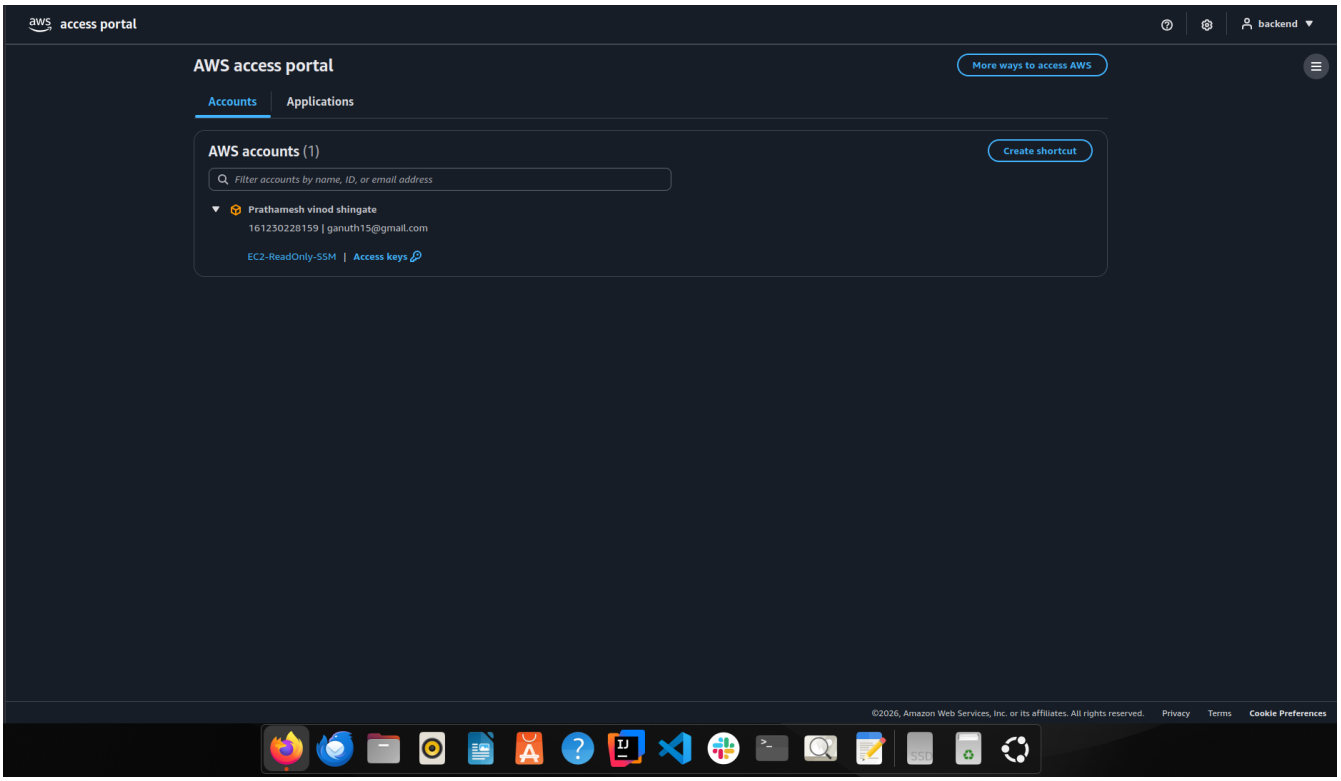
The instance status changed from **Offline to Online**, confirming successful SSM connection.

Step 10: Login Using Identity Center User

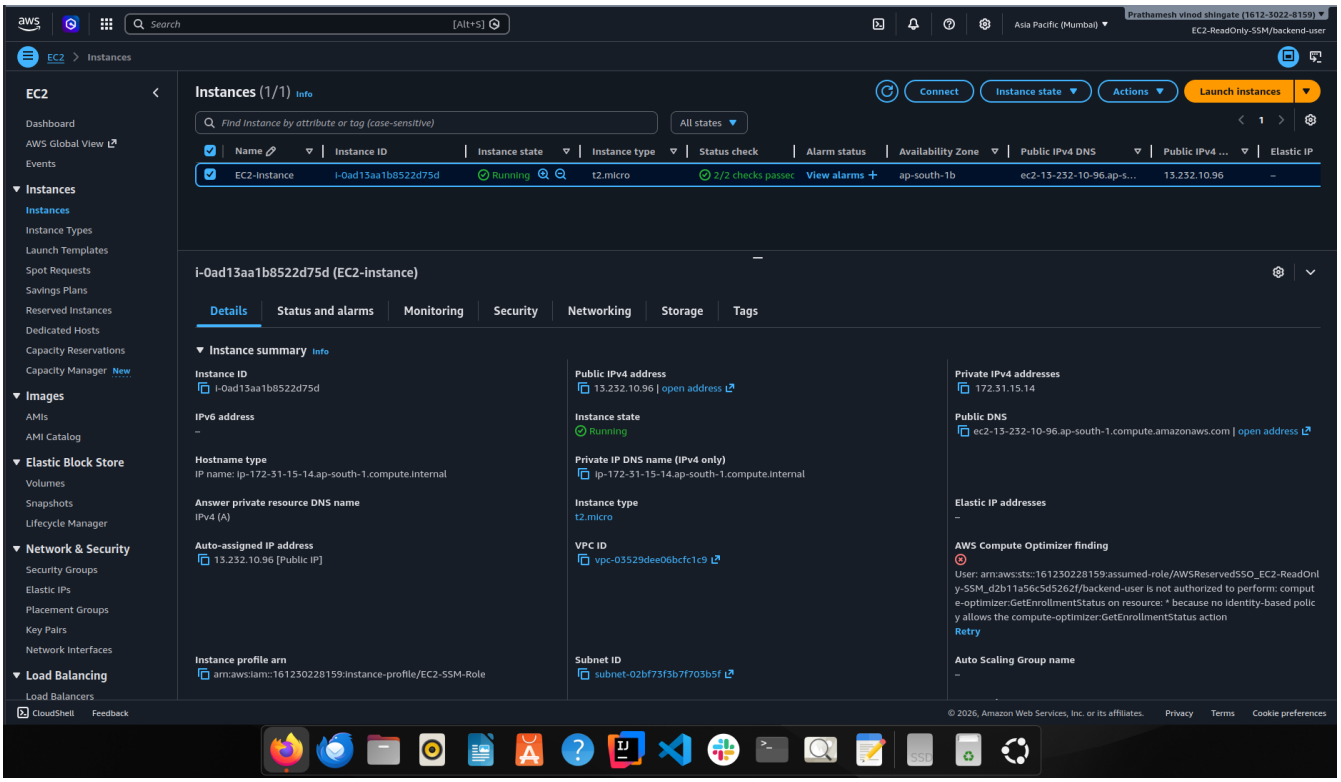
The IAM Identity Center user portal URL was opened.

The Devops-user and backend-user logged in separately using their credentials.

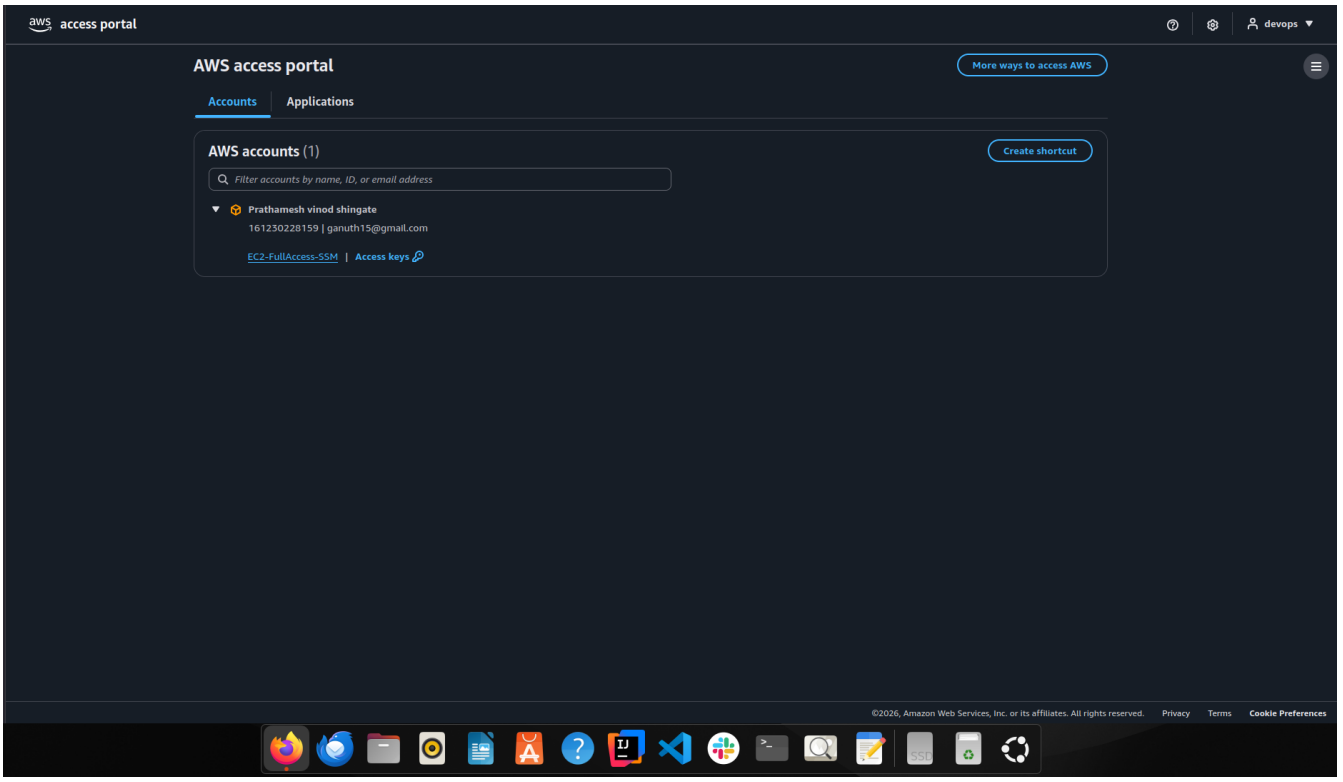
For-**backend-user**



The Backend-user has only EC2-ReadOnlyAccess.



For-Devops-user



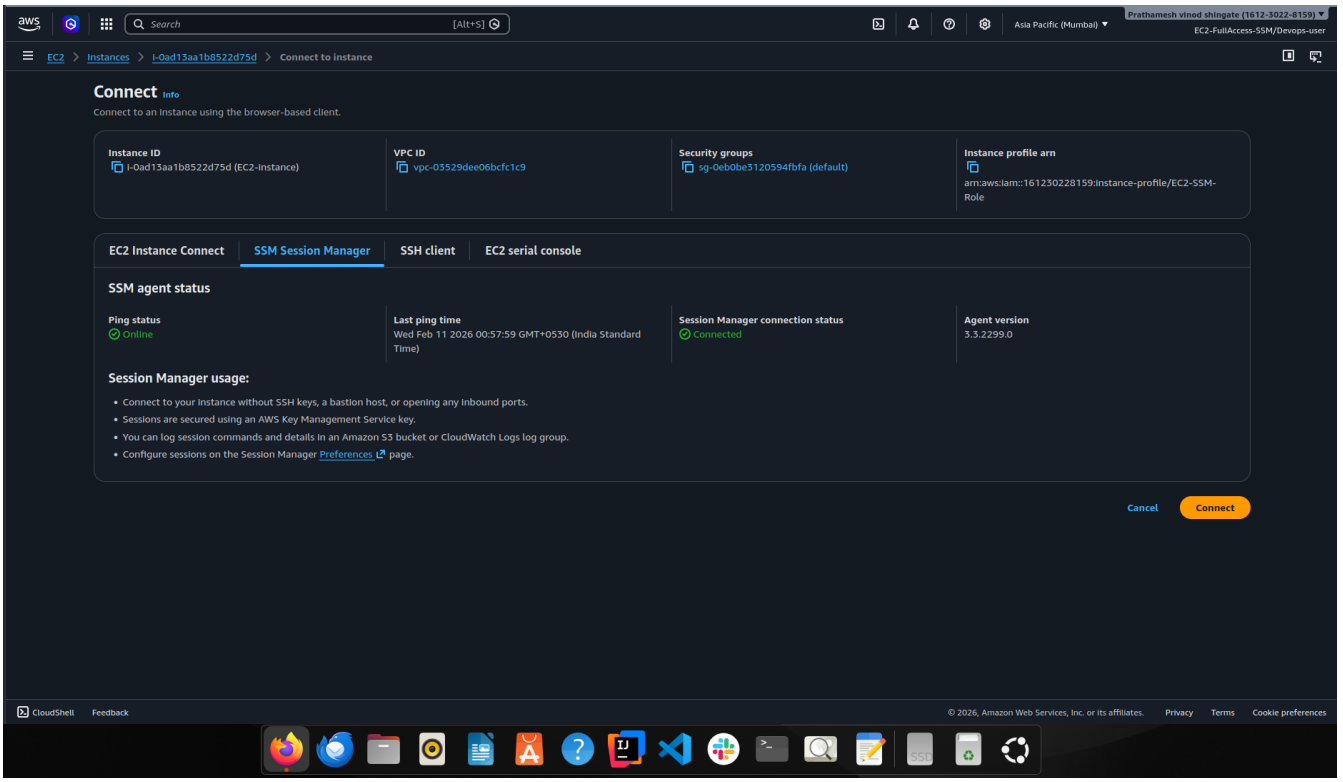
Step 11: Connect to EC2 Using Session Manager

From the EC2 console, the instance was accessed using **Session Manager.**

No SSH key or public IP was required.

The **DevOps user** was able to **start, stop, and connect to the instance.**

The **Backend user** was **only able to view the instance**.



Final Result

IAM-based access to EC2 was successfully configured using IAM Identity Center.

SSH key sharing was avoided.

Access was controlled using permission sets based on user roles.

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